Summaries that tell the truth

Data engineering · Level 2 · Lesson 4



In this lesson you will

- Work out the mean, the median and the range.
- See how one extreme value drags the mean.
- Make an honest chart.

Key ideas

- Mean: add and divide. Median: the middle.
- Extreme values drag the mean, not the median.
- Honest charts start at zero.

Line up, find the middle

1. Collect (3 min): one number each
2. Line up (4 min): find the median
3. Add (4 min): find the mean
4. Change (4 min): one extreme value

Use numbers nobody minds sharing, or let volunteers make them up.

Let’s discuss

1. Why didn’t the median move when one number became huge?
2. When would you report the median instead of the mean?
3. What makes a chart honest?

Exit ticket

- Write five numbers whose mean and median are far apart, and say which one you’d report.